

MYMENSINGH GIRLS’ CADET COLLEGE
PROGRESS TEST EXAMINATION - 2025
CLASS: HSC
MULTIPLE CHOICE QUESTIONS
STATISTICS
SECOND PAPER
[According to the Syllabus of 2026]
TIME – 20 minutes
FULL MARKS – 20

Subject Code:

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Set:

Gha

[N.B. – Answer all the questions. Each question carries ONE mark. Block fully, with a black ball- point pen, the circle of the letter that stands for the correct/best answer in the “Answer sheet” for the Multiple Choice Questions Examination.]

Candidates are asked not to leave any mark or spot on the question paper.

1. How many samples of size 4 can be drawn without replacement from a population of size 10?
(a) 200 (b) 220 (c) 210 (d) 550
2. In sampling with replacement, the probability of each selected sample is –
(a) $\frac{1}{N^n}$ (b) $\frac{1}{N C_n}$ (c) $\frac{1}{N \times n}$ (d) $\frac{n}{N}$
3. Which of the following are true about stratified sampling?
i. The population is divided into homogeneous subgroups called strata
ii. Samples are drawn independently from each stratum
iii. The strata are heterogenous
Which one is correct?
(a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii
4. Which reflectes an unchanging population?
(a) $GRR=1$ (b) $NRR>1$ (c) $GRR<1$ (d) $GRR=NRR$
5. A bacterial culture grows exponentially with a growth rate constant of $r = 0.2$ per hour. How many hours will it take to triple?
(a) $\frac{\ln 3}{0.2}$ (b) $\frac{\ln 2}{0.2}$ (c) $0.2 \times \ln 3$ (d) $0.2 \times \ln 2$
6. Which one is correct?
(a) $TFR>GRR>NRR$ (b) $TFR=GRR=NRR$ (c) $TFR<GRR<NRR$ (d) $NRR>GRR>TFR$
7. If $V(X) = 2$, what is the variance of $(2X + m)$, where m is a constant?
(a) 0 (b) 2 (c) 4 (d) 8
Answer the next TWO questions based on the following information
Let X be a continuous random variable uniformly distributed on the interval $[2, 6]$.
8. What is $E(X)$?
(a) 2 (b) 3 (c) 4 (d) 5
9. What is $V(X)$?
(a) $\frac{4}{3}$ (b) $\frac{8}{3}$ (c) $\frac{16}{3}$ (d) $\frac{5}{3}$
10. If $V(X) = 2$, what is the variance of $(X + 2m)$, where m is a constant?
(a) 0 (b) 2 (c) 4 (d) $2 + 4m$
11. For a discrete random variable X with probability mass function $p(x)$:
i. $P(a \leq X \leq b) = \sum_{x=a}^b p(x)$
ii. $\sum_x p(x) = 1$
iii. $p(x) \geq 0$ for all x
Which one is correct?
(a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

Answer the next three questions based on the following information

X	1	2	3
P(x)	$\frac{1}{3}$	$\frac{1}{6}$	$\frac{1}{2}$

12. What is $P(X \geq 2)$?
- (a) $\frac{1}{3}$ (b) $\frac{1}{2}$ (c) $\frac{2}{3}$ (d) $\frac{5}{6}$
13. Find the cumulative probability $F(2)$.
- (a) $\frac{1}{3}$ (b) $\frac{1}{2}$ (c) $\frac{1}{6}$ (d) $\frac{2}{3}$
14. $P(X = 1 \text{ or } X = 3)$ equals
- (a) $\frac{1}{3}$ (b) $\frac{1}{2}$ (c) $\frac{5}{6}$ (d) $\frac{2}{3}$
15. $P(A \cup B) = P(A) + P(B)$ implies **A & B** are –
- (a) Disjoint (b) Independent (c) Joint (d) Independent

Answer the next THREE questions based on the following information.

A card is drawn at random from a standard deck of 52 playing cards.

16. What is the probability that the card is a Queen?
- (a) 0.0192 (b) 0.0769 (c) 0.25 (d) 0.5
17. P(The card is not a Spade)
- (a) $\frac{1}{4}$ (b) $\frac{1}{2}$ (c) $\frac{3}{4}$ (d) $\frac{1}{3}$
18. P(The card is black or a Heart)
- (a) $\frac{1}{2}$ (b) $\frac{3}{4}$ (c) $\frac{1}{3}$ (d) $\frac{1}{4}$
19. In a class of 40 students, 23 passed in Mathematics, 25 passed in English, and 10 failed in both the subjects. A student is randomly selected. What is the probability that the student passed in at least one subject
- (a) $\frac{1}{3}$ (b) $\frac{1}{4}$ (c) $\frac{3}{4}$ (d) $\frac{4}{5}$
20. $P(A) = \frac{2}{3}, P(B) = \frac{1}{5}$, and $P(A \cap B) = \frac{1}{6}$
- i. A and B are independent
ii. $P(A \cup B) = \frac{7}{10}$
iii. $P(\bar{A}) = \frac{1}{3}$
- Which one is correct?**
- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii